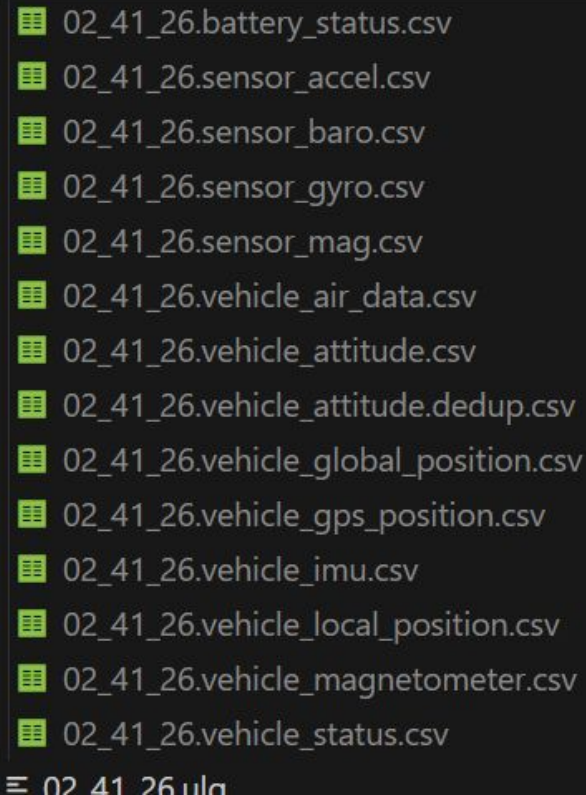


AeroShield Updates

10/22/25

/helpers

- **Ulog_to_csv.py**
 - Takes in .ulg (px4 mission output) and converts it to multiple csv depending on the topics you want
 - Uses the pyulog python library
- **Merge_topics.py**
 - Uses Pandas (python library) to merge all of the csv into one for replay
 - Having issues matching the timestamps due to inconsistent time frequencies in sensor and flight data
- **Csv_to_mavlink_replay.py**
 - This is for relaying the actual mission, using the csv
 - Not working yet, as there are a lot of headers and data that have to match exactly
- An alternative to this is using PX4s built in replay system (might be harder to spoof data, have to do more research)



A screenshot of a file explorer window showing a directory of CSV files. The files are listed in a single column, each preceded by a small green icon representing a document. The files are:

- 02_41_26.battery_status.csv
- 02_41_26.sensor_accel.csv
- 02_41_26.sensor_baro.csv
- 02_41_26.sensor_gyro.csv
- 02_41_26.sensor_mag.csv
- 02_41_26.vehicle_air_data.csv
- 02_41_26.vehicle_attitude.csv
- 02_41_26.vehicle_attitude.dedup.csv
- 02_41_26.vehicle_global_position.csv
- 02_41_26.vehicle_gps_position.csv
- 02_41_26.vehicle_imu.csv
- 02_41_26.vehicle_local_position.csv
- 02_41_26.vehicle_magnetometer.csv
- 02_41_26.vehicle_status.csv

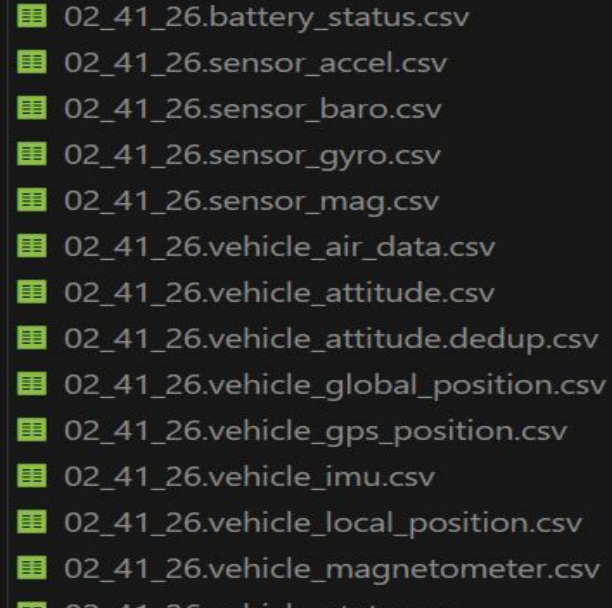
At the bottom of the list, there is a partially visible file named "02_41_26.ulg".

Ulog_to_csv

Takes in .ulg (flight output data from px4), and topics (vehicle_global_position, vehicle_local_position,... theres a lot)

```
python3 ulog_to_csv.py path/to/file.ulg [topic1 topic2 ...]
```

Outputs multiple csvs

A terminal window with a dark background and light green text. It displays a list of 14 CSV files generated by the script, each preceded by a small icon of a document with lines. The files are all named with the prefix '02_41_26.' followed by the topic name.

```
02_41_26.battery_status.csv
02_41_26.sensor_accel.csv
02_41_26.sensor_baro.csv
02_41_26.sensor_gyro.csv
02_41_26.sensor_mag.csv
02_41_26.vehicle_air_data.csv
02_41_26.vehicle_attitude.csv
02_41_26.vehicle_attitude.dedup.csv
02_41_26.vehicle_global_position.csv
02_41_26.vehicle_gps_position.csv
02_41_26.vehicle_imu.csv
02_41_26.vehicle_local_position.csv
02_41_26.vehicle_magnetometer.csv
02_41_26.vehicle_status.csv
```

merge_topics.py

Takes in all csvs that you want to merge, you can specify sampling rate (--hz) which gives the times to merge on, you can also specify the --drop-threshold if you want to get rid of columns with a lot of nan's. This creates two files merged_replay.csv and merged_replay.cleaned.csv

Example use: `python3 merge_topics.py --hz 20 --out merged_replay.csv \`

`02_41_26.vehicle_global_position.csv \`

`02_41_26.vehicle_local_position.csv \`

I have been having problems with matching up the timestamps since some csv like the sensors (imu) data have much more rows than the rest, making it hard to match everything

csv_to_mavlink_replay.py

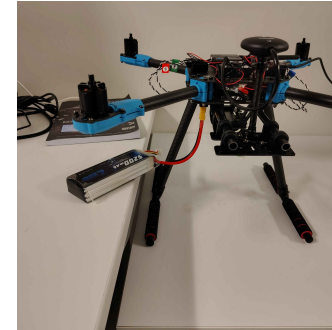
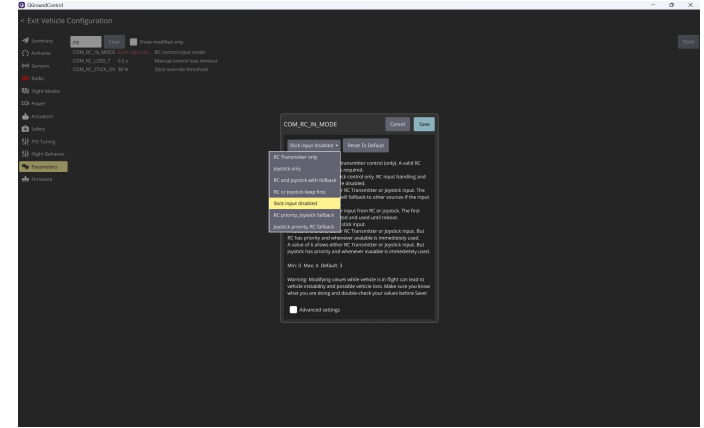
This takes in the new merged csv and replays the mission by sending mavlink messages over udp connection (to QGC for visualization and PX4 for the sensor injection)

Usage: `python3 csv_to_mavlink_replay.py merged_replay.cleaned.csv --target udp:127.0.0.1:14550`

The problem that im having right now is mission or not matching headers properly and not getting the desired visualization

PX4 & QGC Connection to Drone

- 2 Ways of Fixing Connection (Windows)
 - Connect drone to system → Go to QGC Application → “Vehicle Configuration” tab → “Parameters” tab → Search for “joystick” → Select COM_RC_IN_MODE → Go to dropdown and select “Stick input disabled” → Click “Save”
 - Connect drone to system → Go to PX4 terminal → Execute: “commander mode manual”



More drone details

The problem that we were having with connection the drones to QGC is,

First: couldnt get the usb ports to recognize the connection (this is because its running thru WSL, so have to do this, From powershell as admin:

```
usbipd list
```

```
usbipd bind --busid <usb #-#>
```

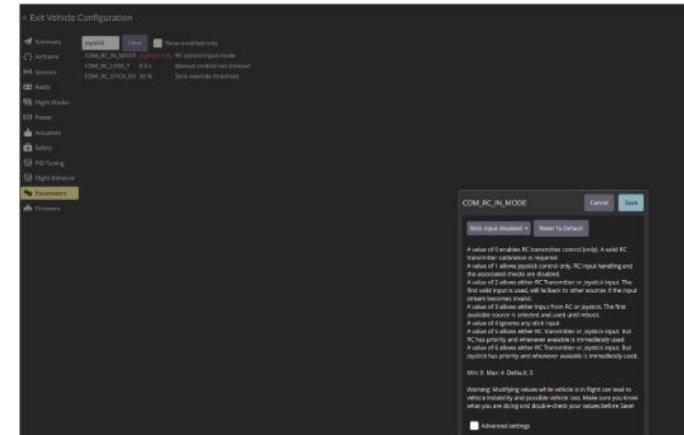
```
usbipd attach --wsl "<ubuntu distro>" --busid <usb #-#>
```

Where usb #-# is the number of the usb you want to share with wsl (from the usbipd list)

More drone details

Second problem was when we did get the connection to wsl to work, QGC would start spam switching between Manual and Hold controls, my thought was that this means that its either looking for a controller to connect to the drone, and when it doesnt find it, it loops looking for it.

So we tried to switch off manual control, so it doesnt look for joystick input anymore, this worked for Dominic and not Zach



Other

- <https://review.px4.io/>
 - Might be good for visualizing and verifying the flight data

PX4 Quadrotor

[Open 3D View](#) [Open PID Analysis](#)

Airframe: Generic Quadcopter
Quadrotor x (4001)
Hardware: PX4_SITL
Software Version: v1.16.0 (RC) (9670eb69)
branch: main
OS Version: Linux, v6.6.87
Estimator: EKF2
Logging Duration: 0:03:33
Vehicle Life
Flight Time: 1 hours 39 minutes 54 seconds

Distance: 219.9 m
Max Altitude Difference: 30 m
Average Speed: 3.7 km/h
Max Speed: 18.3 km/h
Max Speed Horizontal: 18.3 km/h
Max Speed Up: 10.8 km/h
Max Speed Down: 5.5 km/h
Max Tilt Angle: 19.3 deg

Select Some Options

